



Lecture 3: Independence and counting

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1. Review
2. Independence
3. Counting
4. Recitation
5. Summary



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- Conditional probability

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}, \quad \mathbf{P}(B) > 0$$

- Multiplication rule

$$\mathbf{P}(A \cap B) = \mathbf{P}(B) \cdot \mathbf{P}(A \mid B) = \mathbf{P}(A) \cdot \mathbf{P}(B \mid A)$$

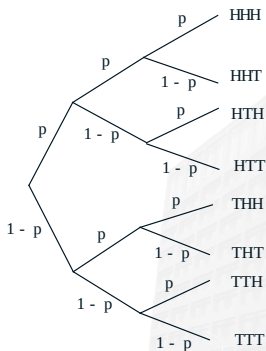
- Total probability theorem

$$\mathbf{P}(B) = \mathbf{P}(A)\mathbf{P}(B \mid A) + \mathbf{P}(A^c)\mathbf{P}(B \mid A^c)$$

- Bayes' rule

$$\mathbf{P}(A_i \mid B) = \frac{\mathbf{P}(A_i)\mathbf{P}(B \mid A_i)}{\mathbf{P}(B)}$$

- 3 tosses of a biased coin: $\mathbf{P}(H) = p$, $\mathbf{P}(T) = 1 - p$
 - $\mathbf{P}(THT) =$
 - $\mathbf{P}(1 \text{ head}) =$
 - $\mathbf{P}(\text{first toss is } H \mid 1 \text{ head}) =$



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Definition of Independence

Events A and B are called **independent** if

$$\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B)$$

or equivalently, when $\mathbf{P}(B) > 0$

$$\mathbf{P}(A \mid B) = \mathbf{P}(A)$$

Remarks:

- “Occurrence of B provides no information about A ’s occurrence”
- Equivalence due to $\mathbf{P}(A \cap B) = \mathbf{P}(B) \cdot \mathbf{P}(A \mid B)$
- Symmetric with respect to A and B
 - applies even if $\mathbf{P}(B) = 0$
 - implies $\mathbf{P}(B \mid A) = \mathbf{P}(B)$ and $\mathbf{P}(A \mid B^c) = \mathbf{P}(A)$
- Does not imply that A and B are disjoint, indeed opposite!
 - Two disjoint events are never independent! (e.g. A and A^c)

- Consider two successive rolls of a 4-sided die.
- Are the events

$$A_i = \{\text{1st roll results in } i\}, \quad B_j = \{\text{2nd roll results in } j\}$$

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Hint:

- We have

$$\mathbf{P}(A_i \cap B_j) = \mathbf{P}(\text{the outcome of the two rolls is } (i, j)) = \frac{1}{16}$$

$$\mathbf{P}(A_i) = \frac{\text{number of elements of } A_i}{\text{total number of possible outcomes}} = \frac{4}{16}$$

$$\mathbf{P}(B_j) = \frac{\text{number of elements of } B_j}{\text{total number of possible outcomes}} = \frac{4}{16}$$

- Consider two successive rolls of a 4-sided die.
- Are the events

$$A = \{\text{1st roll results in 1}\}, \quad B = \{\text{sum of two rolls is 5}\}$$

independent?

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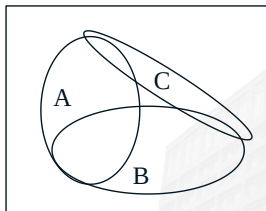
$$A = \{\text{1st roll results in 1}\}, \quad B = \{\text{sum of two rolls is 4}\}$$

independent?

- Conditional independence, given C , is defined as independence under the probability law $\mathbf{P}(\cdot \mid C)$:

$$\mathbf{P}(A \cap B \mid C) = \mathbf{P}(A \mid C) \mathbf{P}(B \mid C)$$

- Assume A and B are independent

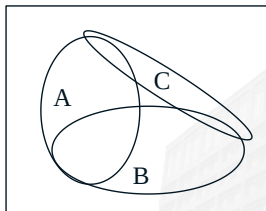


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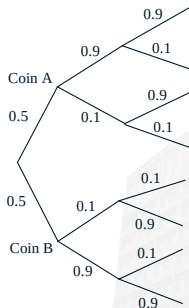


- If we are told that C occurred, are A and B independent?
- Conditioning may affect independence

- Two unfair coins, A and B :

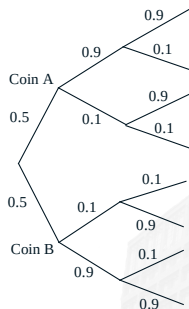
$$\mathbf{P}(H \mid \text{coin } A) = 0.9, \mathbf{P}(H \mid \text{coin } B) = 0.1$$

- We choose either coin with equal probability

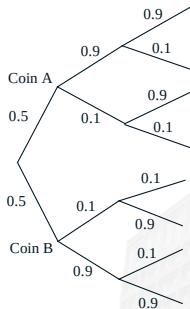


Conditional independence

- Once we know it is coin A , are tosses independent?
- If we do not know which coin it is, are tosses independent?



- Once we know it is coin A , are tosses independent?
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Hint:

- Compare

$\mathbf{P}(\text{11th toss} = H)$ and $\mathbf{P}(\text{11th toss} = H \mid \text{first 10 tosses are heads})$

- **Intuitive definition:** Information on some of the events tells us nothing about probabilities related to the remaining events.
E.g.:

$$\mathbf{P}(A_1 \cap (A_2^c \cup A_3) \mid A_5 \cap A_6^c) = \mathbf{P}(A_1 \cap (A_2^c \cup A_3))$$

Definition: Independence of multiple events

Events $A_1, A_2, \dots, A_n$ are called **independent** if:

$$\mathbf{P}(A_i \cap A_j \cap \dots \cap A_q) = \mathbf{P}(A_i)\mathbf{P}(A_j) \dots \mathbf{P}(A_q)$$

for **any** distinct indices $i, j, \dots, q$ chosen from $\{1, \dots, n\}$.

- Compare with pair-wise independence

Independence vs. pairwise independence

- Two independent fair coin tosses
- A : First toss is H , B : Second toss is H , C : First and second toss give same result

HH	HT
TH	TT

- $\mathbf{P}(C) =$
 - $\mathbf{P}(A \cap C) =$
 - $\mathbf{P}(A \cap B \cap C) =$
 - $\mathbf{P}(C \mid A \cap B) =$
- Pairwise independence **does not** imply independence.

- Let a subsystem consist of components $1, 2, \dots, m$ and let p_i be the probability that component i is up.



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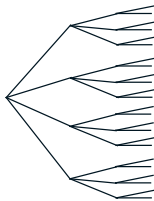
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$$\mathbf{P}(\text{parallel subsystem succeeds}) = 1 - (1 - p_1)(1 - p_2) \cdots (1 - p_m)$$

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- r stages, n_i choices at stage i
 - Number of choices is: $n_1 n_2 \cdots n_r$



- Number of license plates with 3 letters and 4 digits =
 - ... if repetition is prohibited =
- **Permutations:** number of ways of ordering n elements is:
 - Number of subsets of $\{1, \dots, n\}$ =

- Probability that six rolls of a six-sided die all give different numbers?
 - Number of outcomes that make the event happen:
 - Number of elements in the sample space:
 - Answer:



- $\binom{n}{k}$: number of k -element subsets of a given n -element set
- Two ways of constructing an ordered sequence of k **distinct** items:
 - Choose the k items one at a time:

$$n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!} \text{ choices}$$

- Choose k items, then order them ($k!$ possible orders)
- Hence:

$$\binom{n}{k} \cdot k! = \frac{n!}{(n-k)!}$$
$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

- n independent coin tosses, $\mathbf{P}(H) = p$
- $\mathbf{P}(HTTTHHH) =$
- $\mathbf{P}(\text{sequence}) = p^{\# \text{heads}} (1 - p)^{\# \text{tails}}$

$$\begin{aligned}\mathbf{P}(k \text{ heads}) &= \sum_{k\text{-head seq.}} \mathbf{P}(\text{seq.}) \\ &= (\# \text{ of } k\text{-head seqs.}) \cdot p^k (1 - p)^{n-k} \\ &= \binom{n}{k} p^k (1 - p)^{n-k}\end{aligned}$$

- Event B : 3 out of 10 tosses were “heads”.
 - Given that B occurred, what is the (conditional) probability that the first 2 tosses were heads?
- All outcomes in set B are equally likely: probability $p^3(1-p)^7$
 - Conditional probability law is uniform
- Number of outcomes in B :
- Out of the outcomes in B , how many start with HH?

- 52-card deck, dealt to 4 players. Find $\mathbf{P}$ (each gets an ace)

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- Count number of ways of dealing the remaining 48 cards

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- Count number of ways of distributing the four aces: $4 \cdot 3 \cdot 2$
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- Answer:

$$\frac{4 \cdot 3 \cdot 2 \cdot \frac{48!}{12! 12! 12! 12!}}{13! 13! 13! 13!}$$

- Permutations of n objects: $n!$
- k -permutations of n objects: $n!/(n-k)!$
- Combinations of k out of n objects: $\binom{n}{k} = \frac{n!}{(n-k)!k!}$
- Partitions of n objects into r groups, with the i th group having n_i objects:

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1! n_2! \cdots n_r!}$$

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1. About events.

- (a) Can an event A be independent of itself?
- (b) Let A and B be independent events. Use the definition of independence to prove that the events A and B^c are independent.
- (c) Let A , B , and C be independent events, with $\mathbf{P}(C) > 0$. Prove that A and B are conditionally independent of C .

2. Consider two independent fair coin tosses, in which all four possible outcomes are equally likely. Let

$$H_1 = \{\text{1st toss is a head}\},$$

$$H_2 = \{\text{2nd toss is a head}\},$$

$$D = \{\text{the two tosses produced different results}\}.$$

- (a) Are the events H_1 and H_2 (unconditionally) independent?
- (b) Given event D has occurred, are the events H_1 and H_2 (conditionally) independent?

3. Imagine a drunk tightrope walker, in the middle of a really long tightrope, who manages to keep his balance, but takes a step forward with probability p and takes a step back with probability $(1 - p)$.

- (a) What is the probability that after two steps the tightrope walker will be at the same place on the rope?
- (b) What is the probability that after three steps, the tightrope walker will be one step forward from where he began?
- (c) Given that after three steps he has managed to move ahead one step, what is the probability that the first step he took was a step forward?

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- Independence of two events

$$\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B)$$

- Independence of a collection of events

$$\mathbf{P}(A_i \cap A_j \cap \cdots \cap A_q) = \mathbf{P}(A_i)\mathbf{P}(A_j) \cdots \mathbf{P}(A_q)$$

- Counting
- Readings: Section 1.5-1.6
- Assignment: Problem Set 3
- Next lecture: Discrete random variables; Probability mass functions; Expectations.